



CPU EDUCATION

COMP90051

期末班

□ 1 讲解

TUTOR: JJ 学长

时长: 2 小时

本课内容预览:

1. Bias Variance Trade off 与正则化+小题演练

2. PAC learning+大小题演练

最优上界

分解误差



CPU EDUCATION



关于CPU EDUCATION

CPU Education成立于2018年，总部在澳大利亚墨尔本，目前国内办公室在广州。经过近多年的积累与发展，公司凭借优质的教学质量和专业的研发团队力量不断壮大，成为全球最受学生欢迎的教育机构之一，在众多学生及老师中享有良好的口碑。CPU Education争做行业标杆，为不同阶段的学生提供专业，专属，专注的教学服务。

致力于为学生实现人人H1的目标，以“为学生提供极具价值的教学服务，培训各行业精英”为宗旨。秉承“教育高于一切”的经营理念，本着服务定制化、师资专业化、教学规范化的要求，打造墨尔本教育行业的“黄埔军校”。

课后，如果您有任何建议和意见，我们都非常欢迎您联系小助手分享您的想法，给予我们改进和提高的机会！感谢您参与CPU Education的辅导课程！



课程反馈



PART 1 Bias Variance Trade off

Bias-variance decomposition is a fundamental concept for understanding model generalization error.

- Let $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$ denote the training data and x_0 a test point.
- The squared loss is given by:

$$l(Y, \hat{f}_D(x_0)) = (Y - \hat{f}_D(x_0))^2$$

- The expected risk/test error can be decomposed as:

$$E[l(Y, \hat{f}_D(x_0))] = (E[Y] - E[\hat{f}])^2 + \text{Var}[\hat{f}] + \text{Var}[Y]$$

where:

- $(E[Y] - E[\hat{f}])^2$ is the squared bias. 偏差
- $\text{Var}[\hat{f}]$ is the variance. 方差
- $\text{Var}[Y]$ is the irreducible error. 不可约 (这部分误差来自于数据Y观测生成时的噪音, 不可噪音)
- Simple Models:** Typically have high bias but low variance. 欠拟合
- Complex Models:** Typically have low bias but high variance. 过拟合

正则化

2.1 Regularizing Linear Regression

Regularization helps to prevent overfitting and improve model generalization:

- Ridge Regression (L2 Regularization):**

- Adds a penalty term proportional to the squared magnitude of the coefficients.

$$\text{Loss} = \text{MSE} + \lambda \|w\|_2^2$$

- Lasso (L1 Regularization):**

- Adds a penalty term proportional to the absolute magnitude of the coefficients.

$$\text{Loss} = \text{MSE} + \lambda \|w\|_1$$

- Connections to Bayesian MAP:**

- Ridge regression is equivalent to imposing a Gaussian prior.
- X Lasso is equivalent to a Laplace prior.





2.2 Regularizing Non-Linear Regression

Non-linear models often require explicit model selection and regularization:

2.2.1 Explicit Model Selection

- Try different classes of models (e.g., polynomial models of various degrees).
- Use **held-out validation** (cross-validation) to select the model:
 1. Split the training data into training set D_{train} and validation set $D_{validate}$. *未知数据 (unseen data)*
 2. For each model degree d , train model f_d on D_{train} and test on $D_{validate}$.
 3. Pick the degree $\hat{d}$ with the best test score.
 4. Re-train $f_{\hat{d}}$ using all the training data.

多项式
 $d^2 \quad d^3 \quad d^4 \quad \dots d^n$
 $l = 100$

2.2.2 Regularization Objective

- Augment the objective function:

$$\hat{\theta} \in \arg \min_{\theta \in \Theta} (L(data, \theta) + \lambda R(\theta))$$

Weight Decay

(d) Write an expression relating *expected squared loss*, *bias*, and *variance*. [1 mark]

$$E[(Y - \hat{f}(x))^2] = \underbrace{(E(Y) - E(\hat{f}(x)))^2}_{\text{bias}} + \underbrace{\text{Var}(\hat{f}(x))}_{\text{Variance}} + \text{Var}(Y)$$

Expected square loss





- (f) We've seen that the *square-loss risk* of a parameter estimate $\hat{\theta}$ is $\mathbb{E}_\theta[(\theta - \hat{\theta})^2] = [B(\hat{\theta})]^2 + \text{Var}(\hat{\theta})$ while the *square-loss risk* of a supervised regression predictor is $\mathbb{E}_{X,Y}[(Y - \hat{f}(X))^2] = (\mathbb{E}[Y] - [\hat{f}(X)])^2 + \text{Var}(\hat{f}(X)) + \text{Var}[Y]$ where the last term is known as the *irreducible error*. Note that in these risks both $\hat{\theta}$ and $\hat{f}$ implicitly depend on a random training set. Why is there no irreducible error term in the first square-loss risk, even though it appears in the second risk? [5 marks]

在参数估计问题里， θ 是根据 Dataset 估计出来，它虽然与数据随机性相关。但 θ 本身是固定的，所以只有在估计误差（偏差和方差）

但在回归预测中，除了模型本身不确定性， Y 自身还有内在噪声（源自于数据生成观测的），比如即使同一个 X ， Y 在同一个 X 上有波动，这部分是 model 无法消除的

- (b) We saw two *bias-variance decompositions* in the subject: (i) for *parameter estimation* and (ii) for *supervised regression*. Why was there a third *irreducible error* term in case (ii) but not in case (i)? [5 marks]





PART 2 PAC learning

经验误差和 Model Complexity 连接

PAC Inequality

不是绝对的

With probability $\geq 1 - \delta$: $\delta = 0.05$

95% confidence

$$R[f_m] \leq \hat{R}[f_m] + \epsilon(m, \mathcal{F}, \delta)$$

实际风险

经验

Gap

上下界

True Risk of Finite Families

 $\mathcal{F}$ 中 Model 个数

|F| = ? 2, 3, 4

Theorem: Consider any $\delta > 0$ and finite class $\mathcal{F}$. Then w.h.p at least $1 - \delta$:

$$\forall f \in \mathcal{F}, R[f] \leq \hat{R}[f] + \sqrt{\frac{\log |\mathcal{F}| + \log(1/\delta)}{2m}}$$

|F| 无穷

large for simple model

small |F|

small for complex model

large |F|

|F| 大, 第二项大
实际风险上界大

|F| 小 则相反

m 大, 项小
m 小, 项大

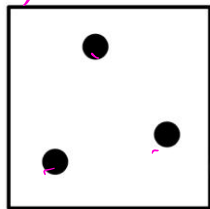
VC Dimension

When $\mathcal{F}$ is Infinite

We need new tools like VC dimension to analyze learnability.

Dichotomies and Growth Function

- **Dichotomies:** Divide the dataset into two disjoint sets (labeling each of them 1 or -1)
- **Growth Function $S_{\mathcal{F}}(m)$:** The max number of label patterns achievable by $\mathcal{F}$ for any m samples

8 ↑
 $S_{\mathcal{F}}(3) = 8$ 

For a model class $\mathcal{F}$ with growth function $S_{\mathcal{F}}(n)$, without knowing the data distribution P , with probability $\geq 1 - \delta$ over the choice of the training set D of n i.i.d. samples

$$R[\hat{f}_D] \leq \hat{R}_D[\hat{f}_D] + 8 \frac{\log S_{\mathcal{F}}(2n) + \log(4/\delta)}{n}$$

Definition of VC Dimension

Definition: The VC dimension $VC(\mathcal{F})$ of a family $\mathcal{F}$ is the largest m such that

Step 1. 先看竹列战

$$S_{\mathcal{F}}(m) = 2^m$$

	x_1	x_2	x_3	x_4
0	0	0	0	0
0	1	1	0	0
1	0	0	0	1
1	1	0	0	1
0	1	0	0	0
1	0	1	0	0
1	1	1	1	1
0	0	0	1	1
0	1	0	0	1
1	1	1	1	0

$$V(\phi) = 4$$
$$\mu(\mathcal{F}) = 3 \quad ?$$
$$\nu(F) = 3$$

4 to 4

0	0	0	✓
0	0	1	✓
0	1	0	✓
1	0	0	✓
0	1	1	✓
1	0	1	✓
1	1	0	✓
1	1	1	✓

Step 2 找
 (x_1, x_2, x_3)

scattered sets

- For a model class $\mathcal{F}$ with VC dimension $\text{VC}(\mathcal{F})$, without knowing the data distribution P , with probability $\geq 1 - \delta$ over the choice of the training set D of n i.i.d. samples

$$R[\hat{f}_D] \leq \hat{R}_D[\hat{f}_D] + \sqrt{8 \frac{\text{VC}(\mathcal{F}) \log(2n+1) + \log(4/\delta)}{n}}$$

当 $V_L(F)$ 过大

$\Delta T_{\text{和}})$ 很大

对于决策没有任何意义





limitation of VC dimension

- Distribution-free and worst-case scenario
- Infinite VC classes may still generalize well
- Does not capture data-dependent complexity
- Difficult to extend to composite functions

VC维入到解我们在最坏情况下
他无法分析在具体分布下
DNN
某些model在特定DNN表现上,
VC无法解释
VC维对复杂情况
难解释

A model family $\mathcal{F}$'s growth function $S_{\mathcal{F}}(m)$ could potentially be 2^m , growing exponentially larger with increasing sample size m . Why would this be bad news for the PAC bound with growth function for $\mathcal{F}$? [5 marks]

PAC 泛化风险上界是为了找一个和泛化风险 Gap 最小的上界

但当 $S_{\mathcal{F}}(m)$ 很大时, Gap 也很大,
这时这个上界无意义的
因为即使在自己的测试集上误差小,
也不能保证泛化误差小

Why might VC-dimension based PAC learning theory lead to very large or impractical risk bounds for deep neural networks with non-linear activations and multiple hidden layers? [5 marks]

DNN 由于多层的结构和非线性激活函数,它能够表达大量函数
所以它的 VC 维大

而 Risk 上界随 VC 维增大而增大
而当 Risk bound, 即使我们地 Error 小, 整体泛化上界也大,
无法评估泛化能力

Probably approximately correct (PAC) learning theory aims to upper bound the true risk $R[f]$ of a learned model $f \in \mathcal{F}$ by the empirical risk $\hat{R}[f]$ of that model plus an error term that might involve a sample size m , VC-dimension $VC(\mathcal{F})$, confidence parameter δ , etc. Why do PAC bounds hold only "with high probability $1 - \delta$ ", and not deterministically? [5 marks]

训练 Data 只是从总体 Data 分布中随机抽样得到的,
所以 $R[f]$ 和 $\hat{R}[f]$ 都具有不确定性
而我们会可能会抽到一些 bad samples 使得上界失效
只能以 $1-\delta$ 概率存在





Since researchers from frequentist statistics have proven that MLE learnings have consistency and efficiency, why do we need PAC theory?

consistency $n \rightarrow \infty$ bias $\rightarrow 0$
efficient $n \rightarrow \infty$ 最小方差

在有限样本, MLE无法告诉我们模型泛化能力是否可靠
且 PAC 还可以定量告诉我们经验风险和泛化实际
风险的差距

What is VC dimension? Why do we need it?

当 $|F|$ 是 infinite, $\log |F|$ 会失效
VC 维来代替使用 $\log |F|$ model complexity

VC 维大起

① Table 的

VC 维

shattered sets

② 证明 VC 维

用于的定义去证明

① 猜 $VC(F)$

② 证 $VC(F) \geq d$

③ 证 $VC(F) \leq d+1$

$VC(F) = d$





The two parts of this question are related to the *VC dimension* but are otherwise unrelated and can be answered separately.

- (a) Consider any binary classifier family $\mathcal{F}$ of classifiers that output labels in $\{0, 1\}$. For any classifier $f \in \mathcal{F}$ define its complementary classifier as $\bar{f} = 1 - f$. That is, for any input x ,

$$\bar{f}(x) = 1 - f(x) = \begin{cases} 0, & \text{if } f(x) = 1 \\ 1, & \text{if } f(x) = 0 \end{cases}$$

Then define a new family $\bar{\mathcal{F}}$ as being made up of all the complementary classifiers of family $\mathcal{F}$. That is, $\bar{\mathcal{F}} = \{\bar{f} : f \in \mathcal{F}\}$. How are $VC(\mathcal{F})$ and $VC(\bar{\mathcal{F}})$ related, and why? [8 marks]

- (b) Consider any finite input domain $X = \{x_1, \dots, x_n\}$ and a family of binary classifiers $\mathcal{F}_{n,k}$ containing all classifiers that output +1 on at most k inputs from X . (That is, this family contains: the classifier outputting all -1; the n classifiers outputting +1 on only one input and otherwise -1; and so on up to classifiers outputting +1 on k inputs and -1 on $n - k$ inputs. What is the VC dimension of this family for general n and $k < n$? Why? [8 marks]

$$VC(\mathcal{F}) = VC(\bar{\mathcal{F}})$$

原因：任意一个点集，如果可以被
某个 $f \in \mathcal{F}$ 实现某种标记组合，那么其互补
也能被 $\bar{f} \in \bar{\mathcal{F}}$ 实现，所以下面的 dichotomy 都能一一实现
例： $f(000) \rightarrow f(111)$ 只是对调了 label
 $\bar{f}(111) \rightarrow \bar{f}(000)$

b). ① 猜：k

$$VC \geq k$$

因为在任意 k 个点，任意子集 可以为 +1，其余 -1
总共能实现 2^k 个 label 组合，所以 $VC(\mathcal{F}) \geq k$

② $VC < k+1$ ，因为 $k+1$ 个点... 全部出 +1 无法被实现
不能 shatter $k+1$ 个点

$$VC = k$$





Question 7: Vapnik-Chervonenkis Dimension [20 marks]

The two parts of this question related to the *VC dimension* but are otherwise unrelated.

- (a) Consider an *input domain* of five points $x_1, \dots, x_5$, and *binary classifier family* $\mathcal{F}$ defined by the table of *dichotomies* given below. Calculate $VC(\mathcal{F})$ and show a corresponding shattered set of points. [6 marks]

x_1	x_2	x_3	x_4	x_5
1	0	1	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
1	1	0	0	0
0	1	1	0	1
0	1	0	1	0
0	1	0	1	1

$\{x_1, x_4, x_5\}$
 $VC(\mathcal{F}) = 3$

- (b) Consider a family $\mathcal{F}$ of classifiers mapping a finite domain of n instances $\{x_1, \dots, x_n\}$ into binary labels $\{-1, +1\}$; and a second family $\mathcal{G}$ of classifiers acting on a *distinct* domain of m instances $\{x_{n+1}, \dots, x_{n+m}\}$. Now define a new *product family* $\mathcal{H}$ that can act on the *combined domain* $\{x_1, \dots, x_{n+m}\}$ where for every pair $f \in \mathcal{F}, g \in \mathcal{G}$, we define an $h \in \mathcal{H}$ such that, for each instance $x_i \in \{x_1, \dots, x_{n+m}\}$, it outputs classification

$$h(x_i) = \begin{cases} f(x_i), & 1 \leq i \leq n \\ g(x_i), & n+1 \leq i \leq n+m \end{cases}$$

What is $VC(\mathcal{H})$ in terms of $VC(\mathcal{F}), VC(\mathcal{G})$? (Hint: It may be helpful to think of families $\mathcal{F}, \mathcal{G}$ as being tables of unique dichotomies in rows, with columns being instances $x_1, \dots, x_n$ and $x_{n+1}, \dots, x_{n+m}$ respectively. Then $\mathcal{H}$ is a table with columns $x_1, \dots, x_{n+m}$.) [14 marks]

先找特殊情形

$$VC(\mathcal{F}) = 3$$

$$VC(\mathcal{G}) = 2$$

f	x_1	x_2	x_3	x_4	x_5
f_1	0	0	0	0	0
f_2	0	0	1	0	0
f_3	0	1	0	0	0
f_4	0	1	1	0	0
f_5	1	0	0	0	0
f_6	1	0	1	0	0
f_7	1	1	0	0	0
f_8	1	1	1	0	0

g	x_{n+1}	x_{n+2}
g_1	0	0
g_2	0	1
g_3	1	0
g_4	1	1

$$f_4 \text{ 和 } g_1 = 01100$$

$$8 \times 4 = 32$$

32 种不同的 label 组合

$$(32) = 2^5 \quad VC = 5 = VC(\mathcal{F}) + VC(\mathcal{G})$$

$$\mathcal{H} \text{ 的 VC 维} \quad 2^m \times 2^n = 2^{m+n}$$

$$\text{所以 } VC(\mathcal{H}) = VC(\mathcal{F}) + VC(\mathcal{G})$$



**Question 4: VC Dimension [14 marks]**

In class we saw that the family of linear classifiers in $\mathbb{R}^2$ (the 2D plane) have Vapnik Chervonenkis (VC) dimension 3. This question asks you to consider *VC dimension* of an unrelated family of classifiers also *in the plane*. Consider the family $\mathcal{F}$ of *rectangle classifiers* parametrised by reals a, b, c, d where $a < b$ and $c < d$: for any instance $\mathbf{x} \in \mathbb{R}^2$ we have the classifier,

$$f_{abcd}(\mathbf{x}) = \begin{cases} +1 & \text{if } a \leq x_1 \leq b \text{ and } c \leq x_2 \leq d \\ -1 & \text{otherwise} \end{cases},$$

or in words, a rectangle classifier predicts +1 if the instance lies within the rectangle, and -1 if the instance falls outside the rectangle.

- (a) What is the VC dimension of $\mathcal{F}$. You do not need to justify your answer in this part—you will do so in the remaining parts. [2 marks]
- (b) If you answered some number d to Part (a), prove that $VC(\mathcal{F}) \geq d$ in this part by drawing d points that can be shattered by $\mathcal{F}$. Demonstrate how each of the 2^d labellings is possible for these points: provide a new drawing for each labelling consisting of the points and a rectangle overlapping some of the points. [10 marks]
- (c) If you answered some number d to Part (a), prove that $VC(\mathcal{F}) < d + 1$ in this part by arguing why there can be no set of $d + 1$ points that can be labelled in all ways by classifiers in $\mathcal{F}$. [2 marks]

Handwritten student work for parts (b) and (c):

Part (b) shows several sets of points and rectangles. The first set has 4 points, and the student has drawn 16 different rectangles, each corresponding to a different labelling of the points. The second set has 5 points, and the student has drawn 16 different rectangles, each corresponding to a different labelling of the points. The third set has 6 points, and the student has drawn 16 different rectangles, each corresponding to a different labelling of the points. The fourth set has 7 points, and the student has drawn 16 different rectangles, each corresponding to a different labelling of the points.

Part (c) shows a set of 5 points. The student has drawn 16 different rectangles, each corresponding to a different labelling of the points. The student has written: "第5个点在四边形内部" (The 5th point is inside the quadrilateral) and "第5个点在四边形外部" (The 5th point is outside the quadrilateral). The student has also written: "2+4+6+4=16" and "2^4=16".



**Question 16: Vapnik-Chervonenkis Dimension [12 marks]**

Consider an *input domain* of six points $x_1, \dots, x_6$, and *binary classifier family* $\mathcal{F}$ defined by the table of *dichotomies* given below.

x_1	x_2	x_3	x_4	x_5	x_6
1	0	1	1	1	0
0	0	0	0	1	1
0	1	0	1	1	0
0	0	1	0	1	1
0	0	1	1	1	0
1	0	0	0	0	1
1	0	0	1	0	0
1	1	1	0	1	1

x_1, x_3, x_4

1	1	1
0	0	0
0	0	1
0	1	0
0	1	0
0	0	0
0	0	0
0	1	0

a) Calculate $VC(\mathcal{F})$ and show a corresponding shattered set of points. [5 marks]

b) List all shattered sets of points. How many are there in total? [7 marks]

c) $VC(\mathcal{F}) = 3$ $\{x_1, x_3, x_4\}$

b). 大小为1: $\{x_1\}, \{x_2\}, \{x_3\}, \{x_4\}, \{x_5\}, \{x_6\}$

大小为2: $\{x_1, x_2\}, \{x_1, x_3\}, \{x_1, x_4\}, \{x_1, x_6\}$

$\{x_2, x_3\}, \{x_2, x_4\}, \{x_2, x_6\}$

$\{x_3, x_4\}, \{x_3, x_6\}$

$\{x_4, x_5\}$

$\{x_5, x_6\}$

大小为3: $\{x_1, x_3, x_4\}, \{x_1, x_3, x_6\}$



**Question 3: Vapnik-Chervonenkis Dimension [16 marks]**

The two parts of this question are related to the *VC dimension* but are otherwise unrelated and can be answered separately.

- (a) Consider an *input domain* of five points $x_1, \dots, x_5$, and *binary classifier family* $\mathcal{F}$ defined by the table of *dichotomies* given below. Calculate $VC(\mathcal{F})$ and show a corresponding shattered set of points. [8 marks]

x_1	x_2	x_3	x_4	x_5
0	0	0	0	1
1	0	1	0	1
0	1	0	1	0
0	1	1	0	1
0	0	0	1	0
1	0	1	1	1
1	1	0	0	1
0	1	1	1	1

x_2, x_3, x_4

0	0	0
0	1	0
1	0	1
1	1	0
0	0	1
0	1	1
1	0	0
1	1	1

$$VC(\mathcal{F}) = 3$$

- (b) For any $t \in [0, 1]$, define the function I_t on $x \in [0, 1]$ to be

$$I_t(x) = \begin{cases} 1, & \text{if } x \geq t \\ 0, & \text{otherwise} \end{cases}$$

What is the *Vapnik-Chervonenkis dimension* of the family $\mathcal{F} = \{I_t : t \in [0, 1]\}$ and why? [4 marks]

- (c) Referring to the functions defined in the previous part, what is the *Vapnik-Chervonenkis dimension* of family $\mathcal{F}' = \mathcal{F} \cup \{1 - I_t : t \in [0, 1]\}$ and why? (Hint: function $1 - I_t$ outputs 0 on inputs $x \geq t$, and 1 otherwise.) [4 marks]

b) 2 个点

+	+
-	+
-	-
+	-

无法实现 $(+1, -1)$
这种 label 组合
 $VC(\mathcal{F}) = 1$

c). 2 个 OK 3

-	+	-
+	+	-
-	-	-
+	-	-

这种不行

$$VC(\mathcal{F}') = 2$$



**Question 3: Vapnik-Chervonenkis Dimension [16 marks]**

The two parts of this question are related to the *Vapnik-Chervonenkis (VC) dimension* but are otherwise unrelated and can be answered separately.

- (a) Consider an *input domain* of five points $x_1, \dots, x_5$, and *binary classifier family* $\mathcal{G}$ defined by the table of *dichotomies* given below. Calculate $VC(\mathcal{G})$ and show a corresponding shattered set of points. [8 marks]

x_1	x_2	x_3	x_4	x_5
0	0	0	0	1
1	0	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	0	1	1
1	0	1	1	1
1	1	1	0	1
0	1	1	1	1

x_3, x_4
0 1
0 1
1 0
1 1

$$VC(\mathcal{G}) = 2$$

- (b) In class we saw that the family of linear classifiers in $\mathbb{R}^2$ (the 2D plane) have *VC dimension* 3. This question asks you to consider *VC dimension* of an unrelated family of classifiers also *in the plane*. Consider the family $\mathcal{F}$ of *rectangle classifiers* parametrised by reals a, b, c, d where $a < b$ and $c < d$: for any instance $\mathbf{x} \in \mathbb{R}^2$ we have the classifier,

$$f_{abcd}(\mathbf{x}) = \begin{cases} +1, & \text{if } a \leq x_1 \leq b \text{ and } c \leq x_2 \leq d \\ -1, & \text{otherwise} \end{cases}, \text{ axis dig h}$$

or in words, a rectangle classifier predicts +1 if the instance lies within the rectangle, and -1 if the instance falls outside the rectangle. What is the *VC dimension* of $\mathcal{F}$, and why? [8 marks]

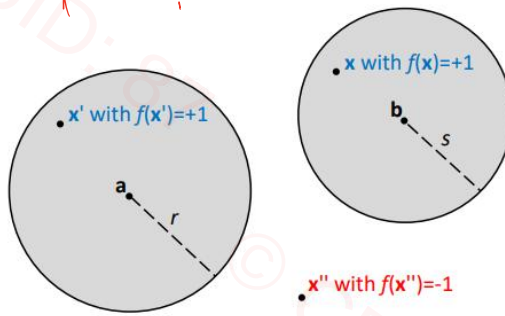
见前面



**Problem III [3 points]**

In this problem, you will further your understanding of Vapnik Chervonenkis (VC) dimension. In class we saw that the family of linear classifiers in $\mathbb{R}^2$ (the 2D plane) has VC dimension 3. This question asks you to consider the VC dimension of an unrelated family of classifiers also *in the plane*.

Consider the family $\mathcal{F}$ of *two-circles classifiers*, where each classifier $f_{a,r,b,s} \in \mathcal{F}$ is parametrised by two distinct circle centers $a, b \in \mathbb{R}^2$ and two positive radii $r, s > 0$. For any instance $x \in \mathbb{R}^2$, $f_{a,r,b,s}(x)$ outputs +1 if x lies within *at least one* of the two circles: the circle with center a and radius r , or the circle with center b and radius s . If x lies outside *both* of the two circles, then the classifier outputs -1. The following figure depicts one such *two-circles classifier* from $\mathcal{F}$ with three input instances demonstrating both classes.



Question 5) [3 points] What is the VC dimension of $\mathcal{F}$? Provide evidence or arguments for your answer.

由于一个圆 VC 维是 3
2 个圆 VC 维应该还是 6

1 +1 -1 交错

所有的
label 组合都有

所有 8 种 label
组

所有 8 种 label
组

8x8 = 64 种 label 组合

= 2⁶

VC(F) ≥ 6

2 个圆 无交集

-1 +1 -1 +1 -1 +1 -1

VC < 7

一个圆只能通过一条不中断的圆弧





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